

DAEGWALLYEONG, Korea, Republic of

WMO: 471000

Lat: 37.6772N

Lon: 128.7183E

Elev: 774

StdP: 92.37

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-15.7	-25.2	0.4	-14.5	-23.4	0.5	-12.8	14.9	-8.9	13.5	-8.0	4.4	270	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	27.3	21.1	25.9	20.5	24.7	19.9	22.6	25.5	21.8	24.4	21.2	23.6	4.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	18.0	24.2	21.0	17.2	23.4	20.3	16.5	22.8	71.1	25.7	68.0	24.5	65.4	23.7	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.7	9.4	-21.1	29.4	2.5	1.1	-22.9	30.2	-24.4	30.8	-25.8	31.4	-27.6	32.3		
(5)			-21.6	23.6	2.4	1.0	-23.4	24.3	-24.8	24.8	-26.1	25.4	-27.9	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-7.0	-4.6	0.4	7.0	12.6	16.2	20.1	20.1	14.8	9.2	2.5	-4.5		
	DBStd	10.07	4.18	4.26	4.34	4.18	3.39	3.01	3.02	2.90	3.04	3.31	4.20	4.39		
	HDD10.0	2089	527	410	297	107	14	1	0	0	2	55	227	449		
	HDD18.3	4192	786	643	554	339	179	75	19	17	113	284	476	707		
	CDD10.0	1102	0	0	1	19	96	187	312	313	145	29	1	0		
	CDD18.3	164	0	0	0	0	2	12	73	72	6	0	0	0		
	CDH23.3	599	0	0	0	3	21	64	242	252	17	0	0	0		
	CDH26.7	63	0	0	0	1	2	5	27	28	1	0	0	0		
(14) Wind	WSAvg	3.8	4.9	4.4	4.4	4.5	4.0	2.9	3.4	2.9	2.4	3.2	4.3	4.9		
(15) Precipitation	PrecAvg	1684	59	51	72	99	108	165	295	380	228	109	76	41		
	PrecMax	2462	244	158	189	245	203	350	697	785	674	417	259	188		
	PrecMin	1145	4	1	14	21	30	25	108	124	1	18	8	6		
	PrecStd	408	65	42	49	54	53	90	154	192	164	110	65	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	9.9	15.3	22.4	26.0	27.2	29.0	29.0	25.3	20.5	15.1	8.2		
		MCWB	1.4	6.3	7.7	12.5	14.3	17.2	22.4	22.4	19.1	13.8	9.5	4.8		
	2%	DB	2.4	6.0	12.4	19.2	23.1	25.1	27.3	27.4	23.1	18.0	12.7	5.5		
		MCWB	-0.3	2.2	6.0	10.0	13.3	17.0	21.6	22.1	18.2	12.2	8.7	2.8		
	5%	DB	0.8	3.6	9.8	16.9	21.2	23.5	25.9	26.0	21.4	16.3	11.0	3.6		
		MCWB	-1.3	0.3	4.6	9.0	12.9	16.7	20.9	21.7	17.6	11.7	7.3	1.0		
	10%	DB	-0.6	1.8	7.6	14.7	19.5	22.1	24.6	24.7	19.9	14.8	9.2	1.9		
		MCWB	-2.5	-1.0	3.1	8.3	12.2	16.6	20.7	21.0	17.0	11.4	6.2	-0.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	7.2	9.2	14.9	17.1	20.5	23.5	23.9	20.9	16.4	11.9	5.6		
		MCDB	3.8	8.0	13.0	19.9	21.0	23.2	26.9	27.0	23.6	17.2	13.2	7.3		
	2%	WB	0.3	2.4	7.0	12.0	15.6	19.3	22.6	22.8	19.5	14.2	9.9	3.2		
		MCDB	1.5	5.0	10.7	17.0	19.6	22.2	25.4	25.9	21.5	16.1	12.1	4.8		
	5%	WB	-1.0	0.7	5.2	10.2	14.5	18.4	21.7	22.1	18.5	13.1	8.1	1.4		
		MCDB	0.4	3.2	9.2	15.1	18.7	21.5	24.3	24.8	20.5	14.8	9.9	3.1		
	10%	WB	-2.3	-0.6	3.4	8.8	13.3	17.5	21.1	21.5	17.4	12.1	6.6	0.0		
		MCDB	-0.9	1.4	7.1	13.5	17.8	20.6	23.7	23.9	19.4	13.9	8.6	1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.2	8.9	10.2	10.1	8.2	5.6	6.0	7.7	9.6	8.7	8.7		
		MCDBR	9.9	10.9	11.9	14.2	13.7	11.8	8.0	8.2	10.3	12.0	11.3	9.8		
	5% WB	MCWBR	7.6	7.7	7.3	7.2	6.2	5.3	3.4	3.8	5.7	6.9	7.3	7.4		
		MCWBR	7.8	9.6	10.8	11.2	10.5	8.5	6.7	7.1	7.9	8.6	8.6	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.369	0.425	0.512	0.549	0.553	0.599	0.566	0.506	0.439	0.431	0.399	0.362		
		taud	2.056	1.935	1.784	1.760	1.792	1.746	1.898	2.071	2.195	2.085	2.096	2.113		
	Ebn at Noon		776	775	743	746	752	717	737	774	808	764	736	754		
		Edh at Noon	123	158	203	220	217	227	194	159	133	134	118	109		
(44) All-Sky Solar Radiation	RadAvg		2.45	3.21	4.11	4.89	5.46	5.13	4.20	4.21	3.88	3.43	2.43	2.19		
	RadStd		0.21	0.31	0.42	0.39	0.51	0.42	0.60	0.55	0.48	0.28	0.31	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.34	N/A	N/A	N/A	+0.53	+0.42	N/A	N/A	+100	N/A
(47) Regional (39 neighbors)		N/A	N/A	-0.87	N/A	N/A	N/A	N/A	N/A	+98	+54

Nomenclature: See separate page